

meeting: "This year, subject to government approval, your company will begin a historic foray into the semiconductor fab and display fab [segment]... We have lined up partners for our semiconductor venture." The company had also released a statement to that effect a few hours after Foxconn's announcement.

However, with Foxconn walking out of the JV, questions have been raised on whether Vedanta still has access to the 40 nm production-grade licence and technology. While the company did not respond to BT's queries, a statement it had released after Foxconn's announcement said, "We will continue to grow our semiconductor team, and we have the licence for production-grade technology for 40 nanometer (nm) from a prominent Integrated Device Manufacturer (IDM)." The JV had resubmitted its application for the incentive scheme based on this licence in June. Further, Rajeev Chandrasekhar, Minister of State for Electronics and IT has told BT that the application was under evaluation.

The company had also said that it was working to acquire a licence for the production-grade 28 nm node.

Be it 28 or 40 nm, whoever becomes the technology partner will have to handhold Vedanta in its semiconductor journey as the technology transfer for chip manufacturing, along with optimising and scaling the yield once production starts, involves numerous steps. "Just because Vedanta gets the technology transfer [deal] doesn't mean that they can use it directly. It could also involve implementing new tools," says independent semiconductor analyst Arun Mampazhy. "In this case, there will be a lot of handholding required for things such as tool-matching, or tweaking. And even if there is a transfer of production technology, yield can only be scaled to 90-100 per cent slowly, not from day one."

But the challenge doesn't just end with finding a technology partner.

A CLEAN BREAK

Vedanta and Foxconn had announced their joint venture in February 2022 with plans to invest \$19.5 billion to set up semiconductor fabrication plant in India

Vedanta, with a 67 per cent share in the JV, was responsible for setting up the fabrication facility, whereas Foxconn, with a 33 per cent share, was responsible for bringing in the technology partner

The Vedanta-Foxconn JV had resubmitted its application based on its licence for 40 nanometre node. The JV was in the process of obtaining technology licence for 28 nm

With Foxconn walking out of the JV, it is not clear if Vedanta still retains the technology licence for the node, while reportedly both the companies are scouting for new partners

With no prior experience in electronics or semiconductor production, any company that partners with Vedanta would have to handhold it in its semiconductor journey

After the wafers are produced, Vedanta would need to find customers for its chips. "Their main problem will be who will be the customer for their wafers," says Satya Gupta, President of VLSI Society. This is something Foxconn, as a manufacturer for leading brands such as Apple, Dell and Xiaomi, among others, could easily have taken care of.

But now, after pulling out of the \$19.5-billion JV, Foxconn hasn't

dropped its plans for setting up semiconductor fabs in India. Instead, reports suggest that Foxconn now intends to set up at least four to five semiconductor fabrication lines in the country, and it has already signed memoranda of understanding (MoUs) with two technology partners. Additionally, the company is also in talks with both Indian as well as global companies for a potential partnership. "It remains to be seen how Foxconn pursues its chip ambitions. It could be another JV with a local conglomerate, or more suitably, with a foreign chip company or foundry it can partner with, to complement its growing manufacturing base in India," says Neil Shah, VP of Research at Counterpoint Technology Market Research.

If industry sources are to be believed, one of the Indian businesses that Foxconn might be in talks with could be the Tata group, that has already announced its plans to venture into semiconductor manufacturing soon. Incidentally, the salt-to-software conglomerate is also in talks to buy Taiwan-based contract electronics manufacturer Wistron's iPhone assembly plant in Karnataka, pitting it against Foxconn—that also manufactures iPhones in India. While the two are likely to compete against each other in everything from smartphones to automotive manufacturing in the coming years, they could join forces in the future to succeed in the chip manufacturing ecosystem in India. Last year, group firm Tata Technologies had joined the Foxconn-led Mobility in Harmony consortium to promote the development of sustainable mobility solutions and encourage collaboration within the industry.

While it is still unclear whether Vedanta and Foxconn will succeed in their plans separately, if they are able to come up with credible technology partners it would prove to be a boost to India's Semiconductor Mission. **BT**

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